

Matthew Quintanilla
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Using AI to Analyze the Physical Outcomes of Viral Infection

In a world dominated by viruses, immense efforts have been made to characterize their infectivity in humans. This allocation of resources and time has led to strides in the field of virology and paved the way for solutions that contribute to global protection, like vaccination. To make these advancements for humankind, researchers use an arsenal of techniques that elucidate information about a virus in a given situation. One of the most rudimentary techniques is titering a virus, which means measuring the amount of virus in a given sample. To do this, researchers infect cells and make observations of the cytopathic effect (CPE) that develops over time. CPE is a physical characteristic that cells exhibit after infection with a virus, and it usually results in the cell's death. By measuring the amount of CPE that a given sample shows over time, researchers can extrapolate the viral titer from the conditions of the infection (i.e., how much volume of virus was added, how many cells are infected, how long it takes for the virus to wipe out a cell population, etc.).

Since each virus uniquely infects cells, and each cell type responds differently to infection, there are also variations in the type of CPE observed during different experimental conditions. Therefore, while tracking the CPE of a cell population after viral infection is vital to research in the field, the technique involves laborious subjective labeling that introduces unwanted variability between the results of any given experiment, and can extend to variability between researchers performing the task. To automate the process and ensure accuracy across experimental trials, researchers have created artificial intelligence (AI) models to detect CPE in different conditions and separately implemented a tracking system that measures the amount of CPE over time to estimate viral titer. These developments could shift the focus from the monotonous tracking of these physical properties to more pressing lab matters, while also reducing the amount of resources needed to analyze CPE data.

Two papers represent the frontier of this research: “A versatile automated pipeline for quantifying virus infectivity by label-free light microscopy and artificial intelligence” and “A web-based artificial intelligence system for label-free virus classification and detection of cytopathic effects.” In the first paper, researchers introduce their AI-powered automated system called the detector of virus-induced cytopathic effect (DVICE), designed to quantify virus titer using label-free light microscopy images interpreted by a convolutional neural network (EfficientNet-B0). Similarly, the second paper introduces an AI model called AI recognition of Viral CPE (AIRVIC), designed to detect and classify virus-induced CPE using a convolutional neural network (ResNet50). Both papers introduce a way for virology researchers to label microscopy images from a virus infection with minimal researcher input.

The DVICE model was trained, validated, and tested on a total of 58,619 microscopy images, 22,873 of which were from infected cells. Their model was first fed labeled microscopy images, and then tasked with identifying features that characterize CPE from each virus and cell type. During their validation phase, they found that DVICE was overfitting training data. To overcome this, they performed leave-one-out cross-validation, which resulted in

a higher area under the receiver operating characteristic (AUROC) curve. Their finalized model was then used to track infected cells that expressed Green Fluorescent Protein (GFP) with live-cell imaging. GFP was only expressed in infected cells, which ensured the model focused on the infected cells for CPE tracking. Based on the detection of GFP, CPE was tracked throughout infection, and viral titers were estimated. Their model was compared to the performance of human annotations, which they found achieved a squared Pearson correlation coefficient of $R^2=0.986$.

While the DVICE model established the groundwork for viral titering through live-cell imaging, the AIRVIC model contributed to a range of virus-specific identification of CPE. With an emphasis on different animal viruses, the AIRVIC model was trained on a total of 40,369 microscopy images (it was not specified whether this dataset was distributed among validation and testing trials). The AIRVIC model learned to identify features distinct to each virus-cell type interaction and was capable of generating classifications for future predictions. During testing, AIRVIC achieved an accuracy of 100% when identifying CPE (whether it was present or not) in MDBK cells infected with the BoGHV-4 DN-599 strain virus, but dropped to a 63.44% accuracy when asked to identify which virus caused infection when the cell line was not specified. To refine their model, they trained AIRVIC for 100 epochs and achieved a stable accuracy for each function they tasked the model with.

DVICE and AIRVIC both make use of convolutional neural networks (CNN) to execute their functions. DVICE used EfficientNetB0, while AIRVIC used ResNet50. EfficientNetB0 is designed to streamline parameters and provide real-time results, which makes it a good candidate for live-cell tracking during viral titering. On the other hand, ResNet50 employs its 50 bottleneck blocks to achieve its high accuracy, but at the cost of time. ResNet50 is, therefore, ideal for the task of high-accuracy CPE identification from microscopy images that can be taken at any point during infection. Both models serve to amplify the efficiency of their given task to alleviate researchers of the tediousness of hand-labeling these often ambiguous images. Titering virus requires cell fixation, staining, and labeling, but DVICE omits the need for these steps. Similarly, in experiments where co-infection may have convoluted the appearance of infected cells, AIRVIC could distinguish between CPE types and reveal which viruses prevail during infection. AI applied to these processes has the capacity to elevate the research done in virology labs that struggle to juggle their experiments. Through this help, life-changing findings can arrive sooner for the world.

References

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